

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 11

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Topic

Evolution of the Reference Parameter Set

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Evolution of the Reference Parameter Set. State all assumptions and conventions.

$$(N_{\theta}, N_O)_{2026} = (47, 52)$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$d^2 = (\theta_{n+1} - \theta_n)^T C_n^{-1} (\theta_{n+1} - \theta_n)$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$L = \sum_{a \in \text{Registry}} R_a^2$$

Show enough intermediate work that another student can reproduce the calculation.